Example. A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains 0.03 kg of salt per liter of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt remains in the tank after half an hour? How much salt is in the tank after a very long time?

1. Let $y(t)$ be the amount of salt (in kg) after t minutes. Give the physical interpretation and the units of $\frac{dy}{dt}$.

2. Use the fact that

$$\frac{dy}{dt} = (\text{rate in}) - (\text{rate out})$$

to find the differential equation satisfied by y.

3. What is the initial condition?

4. Solve the initial-value problem.

5. Find the amount of salt in the tank after 30 min.

6. Find the amount of salt in the tank after a very long time. Does this make sense?

1 Exercises

7. A tank contains 1000 L of brine with 15 kg of dissolved salt. Pure water enters the tank at the rate of 10 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. How much salt is in the tank after 20 minutes? How much salt is in the tank after a long time?

8. The air in a room with volume 180 m^3 contains 0.15% CO_2 initially. Fresher air with only 0.05% CO_2 flows into the room at a rate of 2 m^3/min and the mixed air flows out at the same rate. Find the percentage of carbon dioxide in the room as a function of time. What happens in the long run?

9. A vat with 500 gallons of soda contains 4% sugar (by volume). Soda with 6% sugar is pumped into the vat at a rate of 5 gal/min and the mixture is pumped out at the same rate. What is the percentage of sugar after an hour?

10. A tank contains 1000 L of pure water. Brine that contains 0.05 kg of salt per liter of water enters the tank at a rate of 5 L/min. Brine that contains 0.04 kg of salt per liter of water enters the tank at a rate of 10 L/min. The solution is kept thoroughly mixed and drains from the tank at a rate of 15 L/min. How much salt is in the tank after one hour?